

C. Computation of $\mathbf{K}$ numerically

D. Metropolis acceptance ratio calculation

II. MEASUREMENTS

A related question is if we can use derivative with respect to λ^2 or $(\mathcal{D}(\tau)?)$ to measure $\langle n(\tau)n(\tau') \rangle$

III. LEGENDRE REPRESENTATION

The action for the system is given by an unperturbed term plus a non-local interaction

$$S = S_0 + \iint m(\tau)D(\tau - \tau')m(\tau')d\tau d\tau' \quad (33)$$

The partition function used in QMC can be expressed as a sum over traces and determinants involving a local dynamical weight

$$Z = Z_0 \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \int_0^\beta d\tau_1 \cdots \int_0^\beta d\tau_k \text{Tr}_{\text{loc}} [\dots] \det [\Delta(\tau_i - \tau_j)] W_{\text{loc}}^{\text{dyn}} \quad (34)$$

where $\dots$ refers to the operators as

$$\text{Tr}_{\text{loc}} [\dots] = \text{Tr}_{\text{loc}} [\mathcal{T}_\tau e^{-\beta H_{\text{loc}}} c(\tau_k) c(\tau_{k-1}) \cdots c(\tau_1)] \quad (35)$$

(this is singular expansion in Δ right now, might be missing coefficients for double $\mathcal{J}, \mathcal{U}$ expansion or more concisely

$$Z = \sum \text{Tr}(\dots) \det(\dots) w_{\text{loc}}^{\text{dyn}} \quad (36)$$

This local dynamical weight is defined as an exponential over the kernel K , as in Eq. 8

$$w_{\text{loc}}^{\text{dyn}} = \exp \left(\sum_{\alpha, \beta} S_\alpha S_\beta K_{i(\alpha), j(\beta)}(\tau_\alpha - \tau_\beta) \right) \quad (37)$$

The kernel $K(\tau)$ is determined by solving the differential equation

$$K''(\tau) = D(\tau) \quad (38)$$

subject to the periodic boundary condition $K(0) - K(\beta) = 0$.

We proceed by expanding both $D(\tau)$ and $K(\tau)$ in terms of Legendre polynomials

$$D(\tau) = \sum_n d_n P_n(\tau) \quad (39)$$

$$K(\tau) = \sum_n k_n P_n(\tau) \quad (40)$$

Our goal is to find the transformation matrix $M \in M_{n \times n}(\mathbb{R})$ such that $k = Md$. Setting $\beta = 1$ (WLOG?), the solution for the kernel is obtained by integrating the differential equation

$$K(\tau) = \int_0^\tau (\tau - u) D(u) du - \tau \int_0^1 (1 - u) D(u) du \quad (41)$$

This integral solution can be rewritten using a Green's function

$$K(\tau) = \int_0^1 G(\tau, u) D(u) du \quad (42)$$

where the Green's function is initially identified as

$$G(\tau, u) = (\tau - u)\theta(\tau - u) - \tau(1 - u) \quad (43)$$

By utilizing the identity $\theta(x) + \theta(-x) = 1$, the Green's function can be expressed in a more symmetric form

$$G(\tau, u) = -u(1 - \tau)\theta(\tau - u) - \tau(1 - u)\theta(u - \tau) \quad (44)$$

The matrix elements M_{nq} mapping the Legendre coefficients are thus given by the double integral

$$M_{nq} = \int_0^1 \int_0^1 d\tau du G(\tau, u) P_n(\tau) P_q(u) \quad (45)$$

Substituting the symmetric Green's function splits this into two terms

$$M_{nq} = - \int_0^1 d\tau (1 - \tau) P_n(\tau) \int_0^\tau u P_q(u) du - \int_0^1 d\tau \tau P_n(\tau) \int_\tau^1 (1 - u) P_q(u) du \quad (46)$$

These integrals can be evaluated systematically using the standard recurrence relations for Legendre polynomials

$$x P_n(x) = \frac{(n+1)P_{n+1}(x) + nP_{n-1}(x)}{2n+1} \quad (47)$$

$$\int P_n(x) dx = \frac{P_{n+1}(x) - P_{n-1}(x)}{2n+1} + \text{const} \quad (48)$$

For the periodic case, the partition function is formulated as a path integral

$$Z = \int \mathcal{D}c \mathcal{D}c^\dagger e^{-S(c, c^\dagger)} \quad (49)$$

Differentiating the free energy $F = -\ln Z$ with respect to the expansion coefficients d_n yields

$$\frac{\partial F}{\partial d_n} = \iint \langle n(\tau) n(\tau') P_n(\tau - \tau') \rangle d\tau d\tau' \quad (50)$$

By introducing the correlation function $Q(\tau) = \langle n(\tau) n(0) \rangle = \sum_n q_n P_n(\tau)$, this derivative simplifies to

$$\frac{\partial F}{\partial d_n} = \beta q_n f_n \quad (51)$$

(where q_n are the numbers we are evaluating) Returning to the local dynamical weight, we substitute the Legendre expansion of $K(\tau)$ to find

$$w_{\text{loc}}^{\text{dyn}} = \exp \left(\sum_n k_n \sum_{\alpha, \beta} S_\alpha S_\beta P_n(\tau_\alpha - \tau_\beta) \right) \quad (52)$$

We define a new variable representing the polynomial summation

$$\alpha_n = \sum_{\alpha, \beta} S_\alpha S_\beta P_n(\tau_\alpha - \tau_\beta) \quad (53)$$

Taking the derivative of the free energy with respect to k_n directly computes the expectation value of this variable

$$\frac{\partial F}{\partial k_n} = \langle \alpha_n \rangle \quad (54)$$

Finally, we connect these variations using the chain rule

$$\frac{\partial F}{\partial d_n} = \sum_p \frac{\partial F}{\partial k_p} \frac{\partial k_p}{\partial d_n} \quad (55)$$

From the linear relationship $k = Md$, we know that $\frac{\partial k_p}{\partial d_n} = M_{pn}$. Substituting our previous derivatives into the chain rule relation gives

$$\beta q_n f_n = \sum_p M_{pn} \langle \alpha_p \rangle \quad (56)$$

Solving for q_n isolates the desired correlation coefficients

$$q_n = \frac{1}{\beta f_n} \sum_p M_{pn} \langle \alpha_p \rangle \quad (57)$$